Problem Sheet 3 Solutions

1.

(a) We have $f(t) = P_a(t)$. To find the frequency spectrum of the pulse we use the Fourier transform $F(\omega) = \mathcal{F}\{f(t)\}$:

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t}dt = \int_{-a}^{a} 1.\, e^{-j\omega t}dt$$
$$= -\frac{1}{j\omega}\left[e^{-j\omega a} - e^{j\omega a}\right]$$

Since $e^{-j\omega a} = \cos\omega a - j\sin\omega a$ and $e^{j\omega a} = \cos\omega a + j\sin\omega a$,

$$F(\omega) = -1/j\omega[\cos\omega a - j\sin\omega a - \cos\omega a - j\sin\omega a] = \frac{2j\sin\omega a}{j\omega}$$
$$= \frac{2\sin\omega a}{\omega}.$$

The function ${\sin\omega a}/_{\omega}$ is often written as $sinc_a\omega$.

We can see by inspection that $F(\omega)$ does not go to zero until $\omega \to \pm\infty$. Thus the perfect 'top-hat' shape has associated with it an infinite frequency spectrum.

(b) Such a device would need to generate an infinite frequency spectrum. We could never achieve this as all 'real-world' instrumentation has a finite bandwidth. Consequently, it would be unwise to take the contract!

(c) For $F(\omega \to 0)$ we can either take the standard result $\lim_{x\to 0}\frac{\sin ax}{ax} = 1$ or we can use l'Hopital's rule. Either way

$$F(\omega \to 0) = 2a$$

$F(\omega)$ will first cross the axis at $a\omega = \pm\pi$

$$\omega = \pm\frac{\pi}{a}.$$

(d) As a decreases the pulse in the time-domain gets narrower. In the frequency domain the sinc pulse peak amplitude (at $\omega = 0$) is is reduced while the point at which it crosses the axis is further from the origin, i.e. the spectrum gets flatter and wider. This we anticipate since the narrow pulse requires a wider range of frequencies to support it.

(e) Conversely, as a increases the spectrum becomes taller and narrower. In the limit $a \to \infty$ then $F(0) \to \infty$. The shape starts to resemble the delta function. This we also anticipate since in the limit $f(t) = 1$ for all t and hence we are just taking the Fourier transform of the number 1 and $\mathcal{F}\{1\} = 2\pi\delta(\omega)$. This also makes sense when we consider that $f(t)$ is then a constant (DC input), so the Fourier transform represents a signal at 0 Hz.

Note: it is worth mentioning at this point that an alternative definition of the delta function is

$$\delta(t) = \lim_{a\to 0}\frac{P_{\frac{a}{2}}(t)}{a}$$

It is useful to sketch this function for various a in the range 2…0.25 and check that the area under the pulse is always equal to 1.

Instrumentation

2.

(a) The light intensity is given by

$$f(t)^2 = B^2 e^{-2b^2t^2}$$

where B is an arbitrary constant. The maximum value of $f(t)^2$ is B^2, therefore the time t of the half-maximum is given by

$$0.5 = e^{-2b^2t^2}$$

$$\ln 2 = 2b^2t^2$$

$$t = \frac{1}{b}\sqrt{\frac{\ln 2}{2}}$$

We want the FWHM which is $2t$

$$\Delta t = \frac{2}{b}\sqrt{\frac{\ln 2}{2}}$$

To find $\Delta\omega$ we need the FT of the Gaussian which we can take from the tables of standard transforms to be

$$F(\omega) = \frac{A\sqrt{\pi}}{b} e^{-\frac{\omega^2}{4b^2}}$$

The power density spectrum is therefore

$$F(\omega)^2 = \frac{A^2\pi}{b^2} e^{-\frac{\omega^2}{2b^2}}$$

$$0.5 = e^{-\frac{\omega^2}{2b^2}}$$

$$\ln 2 = \frac{\omega^2}{2b^2}$$

$$\Delta\omega = 2b\sqrt{2\ln 2}$$

So

$$\Delta t\Delta\omega = 4\ln 2$$

$$\Delta t\Delta\nu = \frac{2\ln 2}{\pi} = 0.44$$

(b) For a pulse width $\Delta t = 167$ fs,

$$\Delta\nu = \frac{0.44}{167 \times 10^{-15}} = 2.64 \times 10^{12}\ \text{Hz}$$

(c) For $\lambda_0 = 800$ nm we find the centre frequency

$$\nu_0 = \frac{c}{\lambda_0} = 3.75 \times 10^{14}$$

The pulse occupies a frequency band $\nu_0 - {}^{\Delta\nu}/_2$ to $\nu_0 + {}^{\Delta\nu}/_2$ so

$$\Delta\lambda = \frac{c}{\nu_0 - {}^{\Delta\nu}/_2} - \frac{c}{\nu_0 + {}^{\Delta\nu}/_2}$$

$$= \frac{c\Delta\nu}{\nu_0^2 - \frac{\Delta\nu^2}{4}}$$

Since $\nu_0 \gg \Delta\nu$

$$\Delta\lambda \sim \frac{c\Delta\nu}{\nu_0^2} = 5.6 \text{ nm}$$

3.

(a) For time $t > 0$ Kirchhoff's voltage law give the voltage on the capacitor as

$$V_{out} = V_0 - iR$$

where the current flowing in the circuit is given by

$$i = C\frac{dV_{out}}{dt}$$

hence

$$\frac{dV_{out}}{dt} + \frac{V_{out}}{RC} = \frac{V_0}{RC}$$

Find an integrating factor

$$\mu = e^{\int \frac{dt}{RC}} = e^{\frac{t}{RC}}$$

$$e^{\frac{t}{RC}}\left(\frac{dV_{out}}{dt} + \frac{V_{out}}{RC}\right) = \frac{V_0 e^{\frac{t}{RC}}}{RC}$$

$$\frac{d}{dt}V_{out}e^{\frac{t}{RC}} = \frac{V_0 e^{\frac{t}{RC}}}{RC}$$

$$V_{out}e^{\frac{t}{RC}} = V_0 e^{\frac{t}{RC}} + C$$

Using the initial condition that $V_{out}(0) = 0$ we get $C = -V_0$ so

$$V_{out} = V_0\left(1 - e^{-\frac{t}{RC}}\right)$$

(b) Swap the resistor and the capacitor.

(c) The complex impedance of the capacitor is $Z_C = -j/\omega C$, so treating the filter as a voltage divider we have

$$G = \frac{V_{out}}{V_{in}} = \frac{R}{R - \frac{j}{\omega C}} = \frac{(\omega CR)^2 + j\omega CR}{1 + (\omega CR)^2}$$

(d) The magnitude of the gain is

$$|G| = \frac{\omega CR}{\sqrt{1 + (\omega CR)^2}},$$

which is zero for $\omega = 0$ (i.e. a DC input) and tends to 1 as $\omega \to \infty$, as we would expect for a high-pass filter

(e) The cut-off frequency is by definition where

$$|G| = \frac{1}{\sqrt{2}}$$

Hence,

$$\omega_c = \frac{1}{RC}.$$

(f) For $\omega \ll \omega_c$, we have $|G| \propto \omega$. On a log-log plot this would have a slope of 1; however, recalling that

$$|G|_{dB} = 20\log_{10}|G|,$$

we see that the gain will increase by 20 dB for each decade of frequency (factor of 10 increase in frequency).

4.

(a) A 12-bit ADC will encode $2^{12} = 4096$ digital numbers each representing 1 mV of resolution. So the ADC can digitize signals with amplitude up to 4.096 V.

(b) Dynamic range is range divided by resolution expressed in dB:

$$20\log_{10}\left[\frac{4.096}{10^{-3}}\right] = 72.3 \text{ dB}.$$

(c) The maximum quantisation error is the same size as the resolution, i.e. 1 mV (we could express this as ± 0.5 mV).

(d) RMS quantisation noise added is ${}^{1\text{ mV}}/_{\sqrt{12}} = 0.29$ mV.

5.

(a) Nyquist frequency is ${}^{10^6}/_{2} = 500$ kHz.

(b) Frequencies below the Nyquist limit are properly sampled: 266 and 498 kHz.

(c) 502 kHz will be aliased to $|1000 - 502| = 498$ kHz.1600 kHz will alias to $|1000 - 1600| = 400$ kHz.